

DATA ANALYSIS INTERNSHIP

DAY 20 REPORT

Data Analytics, Inspect Element and CanvasJS

Introduction

Websites such as YouTube and LinkedIn use data analytics extensively to monitor user activities and identify countable patterns.

1.Data Analytics in YouTube

Website:

[YouTube](https://www.youtube.com)

YouTube uses data analytics to analyze user viewing behavior and improve recommendations.

Data Collected:

- Video views
- Likes and comments
- Watch time
- Search history
- Subscriber count

Countable Patterns:

- Trending videos
- Most viewed categories
- User interests

- Peak watching hours

Example:

If many users repeatedly watch cricket highlights, YouTube detects the pattern and recommends similar videos.

2.Data Analytics in LinkedIn



Website:

[LinkedIn](https://www.linkedin.com)

LinkedIn uses data analytics to monitor professional activities and recruitment trends.

Data Collected:

- Profile views
- Job searches
- Skills and endorsements
- Connections
- Post engagement

Countable Patterns:

- Most searched skills
- Hiring trends

- User engagement
- Job application frequency

Example:

If many users search for “Python Developer,” LinkedIn identifies the trend and recommends related jobs.

TASK - CanvasJS Task

Introduction

CanvasJS is a JavaScript library used to create charts and graphs for data visualization. In this task, CanvasJS was used to create and analyze charts such as line charts and pie charts.

Website

[CanvasJS Official Website](#)

Objectives of the Task

- To understand data visualization using CanvasJS
 - To create charts using JavaScript
 - To analyze chart rendering in browser Developer Tools
 - To convert a line chart into a pie chart
 - To verify chart loading using the Network section
-

Uses of CanvasJS

CanvasJS is used for:

- Displaying analytics

- Creating dashboards
 - Visualizing patterns
 - Showing statistical data
-

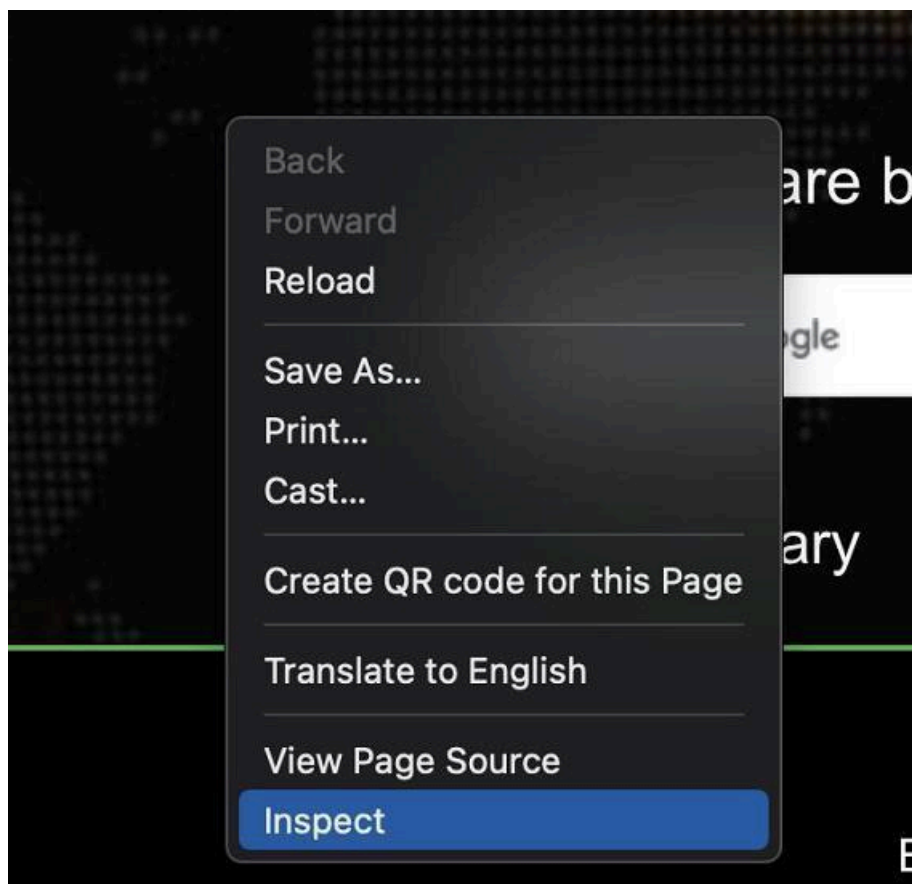
How to Open Inspect Element for CanvasJS

Step 1: Open CanvasJS Website

Open the CanvasJS chart webpage in browser.

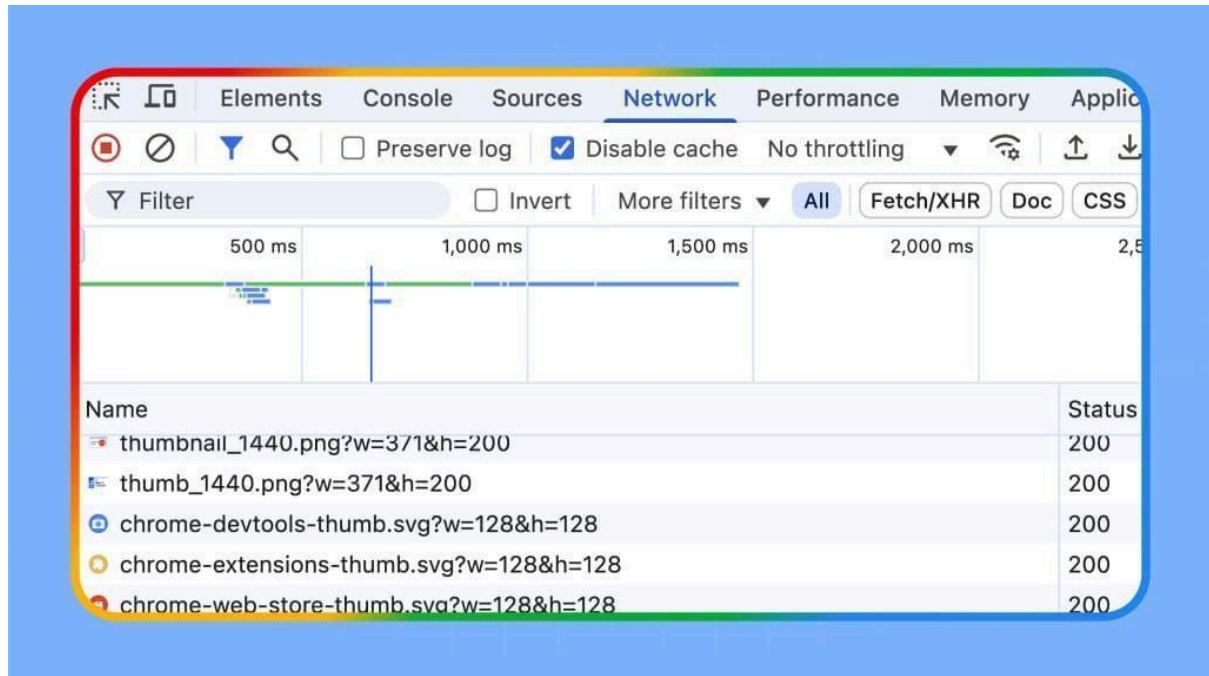
Step 2: Right Click on the Chart Page

Right click anywhere on the webpage containing the CanvasJS chart.



Step 3: Open Network Tab

Click the **Network** tab inside Developer Tools.



Step 4: Refresh the CanvasJS Page

Refresh the webpage to load chart requests and resources.

Step 5: Check CanvasJS Request Status

Find:

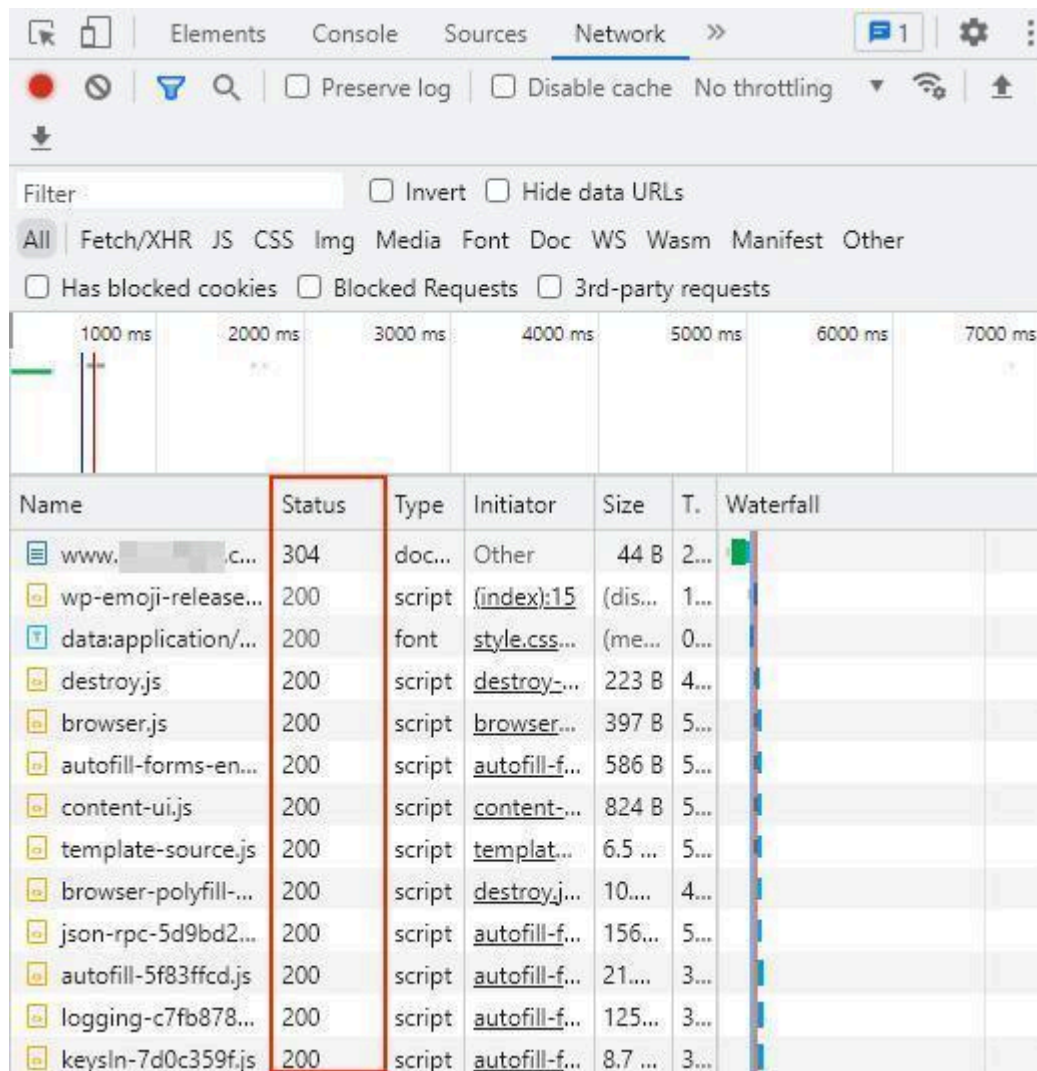
canvasjs.min.js

and check the **Status** column.

Successful Status:

- 200 OK

- 304 Not Modified



Types of Charts Used

1. Line Chart

A line chart displays continuous trends and changes over time.

Example:

type: "line"

Applications:

- Sales growth analysis
 - Website traffic monitoring
 - Student marks analysis
 - Performance tracking
-

2. Pie Chart

A pie chart displays percentage-based data in circular form.

Example:

type: "pie"

Applications:

- Survey analysis
 - Percentage comparison
 - Market share analysis
 - Student result distribution
-

Conversion of Line Chart into Pie Chart

The line chart was converted into a pie chart by modifying the chart type property.

Changed From:

type: "line"

Changed Into:

type: "pie"

```
data: [{  
  type: "pie",
```

This conversion changed the graphical representation from connected lines into circular slices.

Checking Chart Status in Network Section

The Network section in Inspect Element was used to verify whether the chart and resources loaded successfully.

Steps Performed:

1. Opened Inspect Element
2. Opened the Network tab
3. Refreshed the webpage
4. Checked the Status column

Name	Status	Type	In
blob:https://canvasjs.com/3153...	200	docume...	c...
canvasjs.min.js	304	script	bl...
blob:https://canvasjs.com/39e8...	200	docume...	c...
canvasjs.min.js	304	script	bl...
blob:https://canvasjs.com/2980...	200	docume...	c...
canvasjs.min.js	304	script	bl...
blob:https://canvasjs.com/6875...	200	docume...	c...
canvasjs.min.js	304	script	bl...
blob:https://canvasjs.com/cbba...	200	docume...	c...
canvasjs.min.js	304	script	bl...
blob:https://canvasjs.com/b0ad...	200	docume...	c...
canvasjs.min.js	304	script	bl...
blob:https://canvasjs.com/a311...	200	docume...	c...
canvasjs.min.js	304	script	bl...
blob:https://canvasjs.com/bb4e...	200	docume...	c...
canvasjs.min.js	304	script	bl...
blob:https://canvasjs.com/2856...	200	docume...	c...
canvasjs.min.js	304	script	bl...

Status Codes Observed

Status Code	Meaning
200	Request successful
304	Not Modified (loaded from cache)
404	File not found
500	Internal server error

Conclusion

This task helped in understanding how CanvasJS is used for data visualization and chart creation. Different chart types such as line charts and pie charts were analyzed and modified. The Network section in Developer Tools was used to verify successful loading of chart resources using status codes such as 200 and 304.